Section-A

SCENARIO-1:

Question: What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility.

Answer:

When selecting computers for Nexus Innovations, the following factors should be considered:

1. CPU: A modern multi-core processor such as Intel Core i5/i7 or AMD Ryzen 5/7 should be selected. More cores and higher clock speed help the computer handle programming software, web browsers, and video conferencing at the same time.

2. RAM: At least 16 GB RAM is recommended for smooth multitasking. If employees use heavy programming tools or virtual machines, 32 GB RAM can be considered.

3. Storage: A 512 GB or larger SSD is preferred because it provides faster booting, application loading, and file access compared to an HDD. An NVMe SSD can provide even faster performance.

4. Hardware Compatibility: The CPU, RAM, and storage must be compatible with the motherboard. The motherboard should support the CPU socket, RAM type, and SSD interface such as M.2 or SATA.

5. Future Upgrades: The computer should have extra RAM slots and storage options so that the system can be upgraded when the company's requirements increase.

Conclusion: A modern multi-core CPU, 16 GB RAM, and 512 GB SSD would provide good performance for programming, browsing, and video conferencing while ensuring proper hardware compatibility.

SCENARIO-2:

Question: Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

Answer:

Antivirus software continuously monitors the computer to detect and remove viruses, malware, and other threats. While this improves security, it also uses system resources, which can sometimes make the computer slower.

1. CPU Performance:
Antivirus software runs background processes and performs regular security scans. These scans use CPU processing power to check files and programs for threats. During a full system scan, CPU usage can increase, leaving less processing power for other applications.

2. RAM Performance:
Antivirus programs need RAM to keep their services, security tools, and virus definitions running. This reduces the amount of available RAM for other applications. If many programs are open at the same time, the system may become slower or less responsive.

3. Storage Performance:
Antivirus software regularly reads and checks files stored on the computer. During a scan, it creates high storage activity because many files are accessed one by one. This can temporarily slow down file transfers, application loading, and other storage-related operations.

Conclusion:
Antivirus software improves computer security but its background monitoring and scanning consume CPU, RAM, and storage resources. Therefore, systems with limited hardware may experience noticeable slowdowns, especially during full antivirus scans.

SCENARIO-3:

Question: Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

Answer:

If a USB keyboard and printer work properly on one computer but not on another, the problem may be related to driver incompatibility or operating system issues. The following steps can be used to troubleshoot the problem:

1. Check USB Connection:
Connect the keyboard and printer properly and try different USB ports. Also, test the devices on another computer to confirm that the devices themselves are working.

2. Check Device Manager:
Open Device Manager in Windows and look for the keyboard or printer. If there is a yellow warning mark, it may indicate a driver problem.

3. Update Drivers:
Right-click the affected device and select Update Driver. Install the latest compatible driver from the device manufacturer's official website if required.

4. Reinstall the Driver:
If updating does not solve the problem, uninstall the device driver from Device Manager and restart the computer. Windows may automatically reinstall the required driver.

5. Check Windows Updates:
Run Windows Update and install available updates. Some updates contain fixes for hardware compatibility and drivers.

6. Check Compatibility:
Make sure the device driver supports the computer's Windows version and system architecture. An incompatible or outdated driver can prevent the device from working correctly.

7. Restart and Test:
Restart the computer after making changes and reconnect the USB devices. Test the keyboard and printer again.

Conclusion:
By checking the USB connection, Device Manager, drivers, Windows updates, and compatibility, the driver or operating system problem can usually be identified and fixed.

SCENARIO-4:

Question: Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

ANSWER:

Nexus Innovations can use a UPS (Uninterruptible Power Supply) and surge protector to protect computers from power cuts and voltage fluctuations.

1. UPS:
A UPS provides temporary battery power when the electricity goes off. This gives employees enough time to save their work and safely shut down the computer, helping prevent data loss and sudden system shutdowns. It also helps provide a more stable power supply during voltage fluctuations.

2. Surge Protector:
A surge protector protects computer components from sudden increases in voltage, such as power surges. It helps prevent damage to sensitive components such as the motherboard, power supply, storage drive, and other connected devices.

3. Preventing Data Loss:
During a power cut, sudden shutdowns can cause unsaved work to be lost and may sometimes cause file or system corruption. A UPS provides backup power, allowing users to save their work properly before shutting down.

Conclusion:
A UPS protects against power interruptions and helps prevent data loss, while a surge protector protects hardware from sudden voltage surges. Using both provides better protection for office computers.

SCENARIO-5:

Question: Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

ANSWER:

DNS and DHCP are two important network services that help employees' computers communicate properly on the network.

1. DHCP (Dynamic Host Configuration Protocol):
DHCP automatically provides devices with a valid IP address, subnet mask, default gateway, and DNS server address when they connect to the network. If employees sometimes fail to obtain a valid IP address, there may be a problem with the DHCP server or its configuration.

2. DNS (Domain Name System):
DNS converts domain or host names into IP addresses. For example, instead of accessing an application server using 192.168.1.10, employees can access it using its hostname. If the server can be reached by IP address but not by name, it indicates a **DNS problem**.

3. How They Work Together:
DHCP gives the client its network configuration, including the address of the DNS server. The client then uses DNS to resolve server names into IP addresses.

Conclusion:
The symptom “IP address works but name does not” most strongly indicates a DNS misconfiguration. However, the separate symptom of devices sometimes not receiving a valid IP address indicates a DHCP issue as well. Therefore, both DNS and DHCP should be checked, especially the DHCP-provided DNS server settings.

SCENARIO-6:

Question: Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

ANSWER:

After installing a new firewall, employees can access the internet but cannot reach the internal application server. This suggests that the firewall may be blocking legitimate internal traffic. The following steps can be used to troubleshoot and resolve the issue:

1. Confirm the Problem

First, check whether the application server is actually running and reachable from the internal network. Try to ping the server and test the application using its IP address and hostname. This helps determine whether the problem is with the server or the firewall.

2. Check the Firewall Rules

Log in to the firewall management interface and review the configured rules. Check whether there is any rule that blocks traffic from the employees' network to the application server.

Important details to check include:

- **Source IP/network** – employees' network
- **Destination IP** – application server
- **Protocol** – TCP or UDP
- **Destination port** – the port used by the application, such as HTTP/HTTPS
- **Action** – Allow or Deny
- **Rule order** – whether a deny rule is being applied before the required allow rule

3. Check Firewall Logs

Review the firewall's traffic or security logs while an employee tries to access the application server. If the connection appears as Denied, Dropped, or Blocked, the logs can help identify which firewall rule is causing the problem.

4. Verify the Required Port

Determine which port the application server uses. For example, a web-based application may use TCP port 443 for HTTPS. Make sure the firewall allows the required protocol and port between the employee network and the application server.

5. Check Routing and Network Configuration

If the firewall rule appears correct, verify that the firewall has the correct routes to the internal application server and that the server's default gateway points to the correct network device.

6. Create or Correct the Allow Rule

If a firewall rule is blocking legitimate traffic, create or modify an allow rule that permits only the required traffic from the authorized employee network to the application server.

For example:

Source: Employee network
Destination: Application server
Protocol: TCP
Port: Application's required port
Action: Allow

The rule should be placed in the correct order so that it is processed before a general blocking rule.

7. Test the Connection

After making the change, test the application again from an employee computer. Check whether the application loads correctly and review the firewall logs to confirm that the traffic is now being allowed.

8. Document the Changes

Finally, record the firewall rule that was changed, the reason for the change, and the testing results. This helps the IT team troubleshoot similar problems in the future.

Conclusion

The troubleshooting process should involve testing connectivity, checking firewall rules and logs, verifying ports and routing, correcting the required allow rule, and testing again. The goal is to allow only the legitimate traffic required to access the internal application server while keeping unnecessary traffic blocked.

SECTION-B

Task 1: Workstation Assembly

Installation Sequence

Step 1: Prepare the Workspace
I switched off and unplugged all equipment and prepared a clean, dry workspace. I used the PC tool kit and took necessary precautions to avoid static electricity.

Step 2: Install the CPU
I placed the motherboard on a safe surface and opened the CPU socket. I aligned the CPU with the marking on the socket and carefully placed it into the socket. I locked the CPU socket properly.

Step 3: Install the CPU Cooler
I applied thermal paste if it was not already applied to the cooler. Then I positioned the CPU cooler over the processor and secured it properly. I connected the CPU fan cable to the motherboard's CPU_FAN connector.

Step 4: Install RAM
I opened the RAM slot clips and aligned the notch on the RAM module with the notch in the slot. I pressed the RAM firmly until the clips locked into place. If multiple RAM modules were used, they were installed in the recommended motherboard slots.

Step 5: Install the Motherboard into the Chassis
I installed the required I/O shield if applicable and placed the motherboard inside the chassis. I aligned the motherboard with the standoffs and secured it using the appropriate screws.

Step 6: Install Storage Device
I installed the SSD/HDD in its appropriate mounting location. For a SATA drive, I connected the SATA data cable between the drive and motherboard and connected the SATA power cable from the power supply.

Step 7: Install the Power Supply
I placed the SMPS/PSU into the chassis and secured it with screws. I connected the main 24-pin ATX power connector to the motherboard and the 4/8-pin CPU power connector near the processor.

Step 8: Install Other Components
If required, I installed additional components such as a graphics card into the

appropriate PCIe slot and secured it to the chassis. Any required PCIe power cable was also connected.

Step 9: Connect Front-Panel Cables
I connected the chassis front-panel cables to the motherboard, including:

- Power switch
- Reset switch
- Power LED
- HDD/SSD activity LED
- Front USB
- Front audio

I followed the motherboard manual for the correct connector positions.

Step 10: Check Internal Connections
I checked that all required connections were properly attached:

- 24-pin motherboard power
- CPU power
- CPU fan
- RAM
- Storage data cable
- Storage power cable
- Front-panel connectors
- Graphics card and its power cable, if applicable

I also ensured that cables were arranged properly and were not interfering with any fans.

Step 11: Connect External Devices
I connected the monitor, keyboard, mouse, and AC power cable to the workstation.

Step 12: Power On and Perform POST Test
I switched on the power supply and pressed the power button. The system started successfully and performed the Power-On Self-Test (POST). The display showed the startup/BIOS screen, confirming that the basic hardware components were detected.

POST Verification

The workstation successfully passed POST. The CPU, RAM, and storage were detected by the BIOS, and there were no abnormal beep/error indications. Therefore, the hardware assembly was considered successful.

Conclusion

The workstation was successfully assembled by installing the CPU, CPU cooler, RAM, motherboard, storage, power supply, and other required components. All necessary internal and front-panel cables were connected correctly. Finally, the system was powered on and successfully passed POST.

Task 2: RAM Mismatch Troubleshooting

Answer:

Step 1: Power On the Workstation

I connected the workstation to power and switched it on. During startup, I observed the POST screen for any RAM-related warning or error message.

Step 2: Enter BIOS/UEFI

During startup, I pressed the appropriate BIOS/UEFI key, such as Delete, F2, or F10, depending on the motherboard.

I opened the Memory/RAM information section and checked the installed RAM modules.

Step 3: Identify the RAM Mismatch

I compared the specifications of the installed RAM modules, including:

- RAM capacity
- RAM speed/frequency
- RAM timings
- RAM type
- Number of installed modules

For example, if one module is 8 GB DDR4 3200 MHz and another is 8 GB DDR4 2666 MHz, the modules have different operating speeds.

The BIOS/UEFI may show a warning or configure the memory at a lower compatible speed.

Step 4: Determine the Root Cause

After checking the RAM information, I determined that the mismatch was caused by different RAM speeds.

Example root cause:

One RAM module was rated at 3200 MHz while the other module was rated at 2666 MHz. Therefore, the installed modules had mismatched memory speeds.

If your practical shows a different problem, write the actual cause instead:

- **Speed mismatch:** RAM modules have different MHz ratings.
- **Timing mismatch:** RAM modules have different CAS latency/timing values.

- **Capacity mismatch:** RAM modules have different capacities, such as 8 GB + 16 GB.

Step 5: Power Off the System

I shut down the workstation completely and disconnected the power cable. I also took appropriate anti-static precautions before handling the RAM.

Step 6: Reseat or Reconfigure the RAM

I removed and reseated the RAM modules carefully.

I checked that:

- The RAM was installed in the correct motherboard slots.
- The notch on each RAM module was aligned correctly.
- Both modules were firmly locked into their slots.

If required, I used compatible RAM modules with matching specifications.

Alternatively, I entered BIOS/UEFI and configured the memory settings to a common supported speed so that both modules could operate correctly.

Step 7: Save BIOS/UEFI Settings

I saved the changed settings using Save & Exit and restarted the workstation.

Step 8: Verify the RAM

After restarting, I entered BIOS/UEFI again and checked that:

- Both RAM modules were detected.
- The total RAM capacity was correct.
- The memory was operating at a supported/common speed.
- No RAM mismatch warning was displayed.

Step 9: Confirm Normal Boot

I allowed the workstation to continue booting. The operating system loaded normally without displaying the previous RAM mismatch error.

Result

Before Fix:
The BIOS/UEFI detected a RAM mismatch caused by different memory specifications.

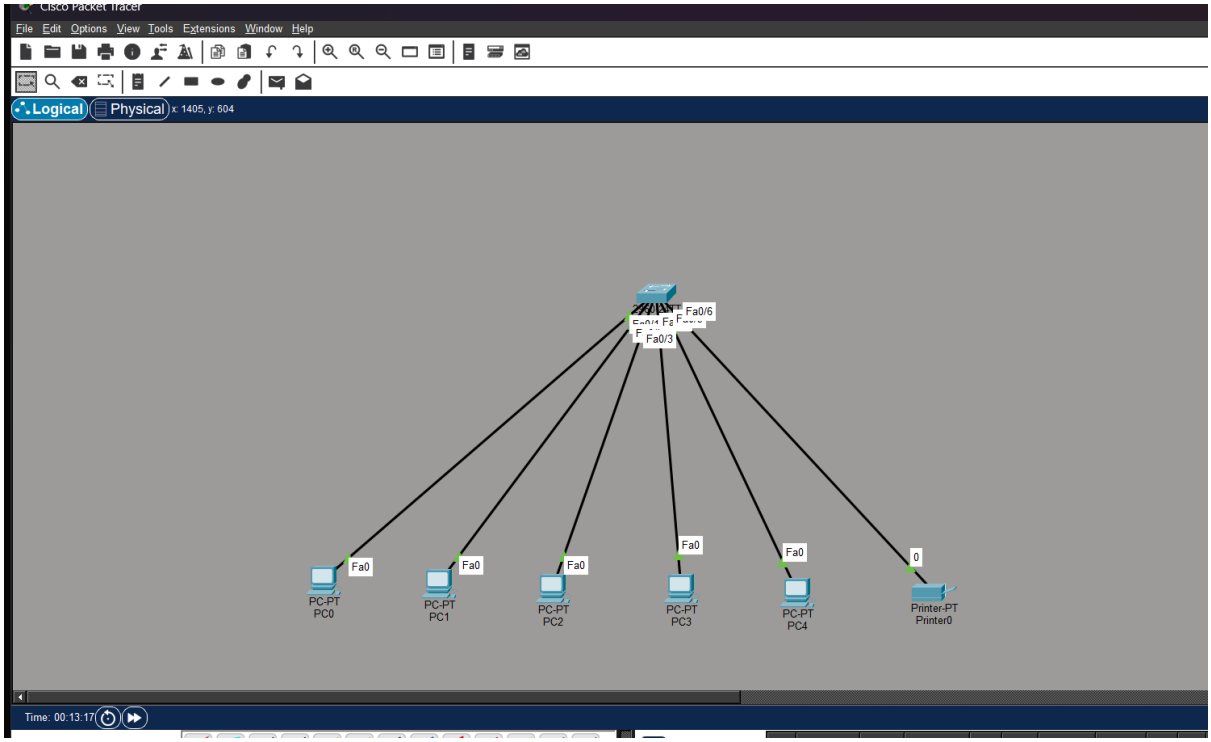
Root Cause:
Mismatch in RAM speed/timing/capacity.

Action Taken:
The RAM modules were reseated and/or the BIOS/UEFI memory configuration was adjusted to use compatible settings.

After Fix:
Both RAM modules were detected correctly, the RAM settings were compatible, and the workstation successfully booted normally.

Task 3: LAN Topology Build

Answer:




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

I created a LAN topology in Cisco Packet Tracer using 5 PCs, 1 network printer, and 1 Cisco 2960 switch. All devices were connected to the switch using Copper Straight-Through cables.

I used the Class C network 192.168.10.0/24 with subnet mask 255.255.255.0. The IP addresses were assigned as follows:

- PC0 – 192.168.10.1
- PC1 – 192.168.10.2
- PC2 – 192.168.10.3
- PC3 – 192.168.10.4
- PC4 – 192.168.10.5
- Printer – 192.168.10.6

The same subnet mask 255.255.255.0 was configured on all devices. Since all devices are on the same LAN, no router or default gateway was required.

Finally, I tested connectivity using the ping command from PC0 to the other PCs and printer. All devices returned successful replies, confirming that the LAN topology and IP configuration were working correctly.

Result: The 5 PCs and 1 printer were successfully connected and all assigned IP addresses were reachable through ping.

Task 4: Clean OS Installation

Objective

The objective of this task was to perform a clean installation of a Windows/Linux operating system using an ISO file, manually partition the disk, and optimize startup programs for faster booting.

Procedure

First, I started the workstation using the provided Windows/Linux ISO file. I selected the required language, keyboard, and installation settings and proceeded with the OS installation.

During disk setup, I selected the Custom/Manual Installation option instead of automatic partitioning. I manually created and formatted the required disk partitions and selected the appropriate partition for installing the operating system.

After completing the partitioning, I continued the installation process. The required system files were copied to the selected partition, and the workstation restarted several times. After installation, the system successfully opened the newly installed operating system.

Next, I checked the Startup Apps/Task Manager and disabled unnecessary programs that were configured to start automatically. Essential system and security services were kept enabled to avoid affecting normal operation.

Finally, I restarted the workstation to verify the changes. The operating system booted normally without displaying any installation or startup errors.

Result

The operating system was successfully installed from the ISO using manual disk partitioning. Unnecessary startup programs were disabled to improve startup performance, and the workstation successfully booted into the OS without errors.

SECTION-C

Nexus Innovations – Diagnosing Random System Reboots

Objective

At Nexus Innovations, three rendering workstations restart randomly during heavy afternoon workloads even though they successfully pass POST every morning. The main factors are likely related to thermal conditions, power delivery, and heavy hardware load.

1. Three Hardware Components Most Likely to Cause Random Reboots

1. CPU (Processor):
Rendering creates high CPU usage and produces significant heat. If the CPU cooler is dusty, improperly mounted, or the thermal paste has degraded, the CPU temperature can become too high. The system may automatically restart to protect the processor.

2. GPU (Graphics Processing Unit):
Rendering workloads heavily use the GPU. A GPU under continuous load can generate a large amount of heat and consume substantial power. Overheating or an unstable graphics card can cause system crashes or sudden restarts.

3. PSU (Power Supply Unit):
A rendering workstation can draw much more power under heavy CPU and GPU load than during normal use. An underpowered, aging, or faulty PSU may fail to provide stable power, resulting in sudden system restarts.

2. Why the Systems Pass POST in the Morning but Fail in the Afternoon

POST mainly checks whether the basic hardware is functioning during startup. In the morning, the computers are relatively cool and are not under heavy workloads, so they can pass POST normally.

During the afternoon, the workstations are operating inside poorly ventilated desk cabinets. Heat builds up around the systems and the internal temperature increases. When rendering starts, the CPU and GPU generate even more heat and demand more power. If temperatures exceed safe operating limits or the PSU cannot provide stable power under the increased load, the system may restart unexpectedly.

Therefore, the difference is mainly caused by increased temperature and power demand during sustained workload rather than a failure that occurs during the initial POST.

3. Using a Stress-Test Tool to Confirm the Problem

I would use a monitoring and stress-testing tool such as OCCT.

First, I would monitor the CPU temperature, GPU temperature, system temperature, and PSU voltage readings at idle.

Next, I would run a CPU stress test and then a GPU stress test. I would also use a combined power test if supported by the tool. During the test, I would monitor temperatures and system stability.

- If temperatures rise rapidly to unsafe levels and the workstation restarts, this would indicate a thermal problem.
- If temperatures remain acceptable but the system restarts when CPU and GPU are loaded together, a power/PSU problem becomes more likely.
- Event Viewer and hardware monitoring logs can also be checked after the restart for additional evidence.

4. Three Specific Remediation Steps

Step 1 – Improve Cooling and Airflow
Remove the workstations from the poorly ventilated desk cabinets or provide proper ventilation around them. Clean dust from the CPU cooler, GPU, fans, and air vents. Ensure that the case fans are working correctly and that airflow is unobstructed.

Step 2 – Check and Maintain CPU/GPU Cooling
Inspect the CPU cooler and replace thermal paste if required. Check that the CPU and GPU fans operate properly and verify temperatures during rendering. Replace a faulty cooler or fan if necessary.

Step 3 – Test and Correct Power Delivery
Check the PSU's wattage, condition, connectors, and voltage stability. Make sure the PSU is sufficient for the combined CPU and GPU power requirements. Replace the PSU if testing shows that it is defective or underpowered.

5. Is a PSU Upgrade Necessary?

A PSU upgrade may not be necessary in every case. The random restarts could be caused mainly by overheating due to poor ventilation. Therefore, the PSU should first be tested under load and its capacity should be compared with the workstation's actual CPU and GPU power requirements.

If the existing PSU has sufficient wattage and provides stable power while temperatures are high, improving cooling may solve the problem without replacing the PSU. However, if the PSU is underpowered, aging, faulty, or voltage becomes unstable during combined

CPU/GPU stress testing, upgrading to a suitable higher-quality PSU would be technically justified.

Conclusion

The most likely causes of the random afternoon restarts are CPU/GPU overheating and possible PSU instability under heavy rendering loads. The poorly ventilated cabinets increase heat buildup, while rendering increases both thermal output and power consumption. Stress testing and monitoring can help identify the exact cause. Improving airflow, maintaining cooling components, and verifying PSU capacity and stability should resolve the frequent system restarts.

Section D

AI-Augmented Learning

Step 1 Build with AI

I used ChatGPT to create an IP addressing and subnetting plan for Nexus Innovations' startup office network.

Exact Prompt Used:

“Create an IP addressing and subnetting plan for Nexus Innovations' startup office network for 5 systems and 1 printer. Use a private Class C IPv4 network. Specify the subnet, subnet mask, usable host range, and broadcast address. Assign a unique IP address to all 5 systems and the printer. Also identify which addresses should be reserved for future expansion.”

AI-Generated Plan:

- **Subnet:** 192.168.10.0/28
- **Subnet Mask:** 255.255.255.240
- **Usable Host Range:** 192.168.10.1 – 192.168.10.14
- **Broadcast Address:** 192.168.10.15

IP Allocation:

- System 1 – 192.168.10.1
- System 2 – 192.168.10.2
- System 3 – 192.168.10.3
- System 4 – 192.168.10.4
- System 5 – 192.168.10.5
- Printer – 192.168.10.6

Reserved for Future Expansion:
192.168.10.7 – 192.168.10.14

Step 2 – Test & Debug Without AI

I manually verified the subnetting plan without using AI. A /28 subnet provides 16 total addresses and 14 usable host addresses. The network address is 192.168.10.0 and the broadcast address is 192.168.10.15. The assigned addresses from .1 to .6 are unique and do not overlap. The remaining addresses from .7 to .14 are available for future expansion.

The plan provides enough addresses for the 5 systems and 1 printer, while also keeping additional addresses reserved for future devices. After verification, the subnetting plan was found to be correct.